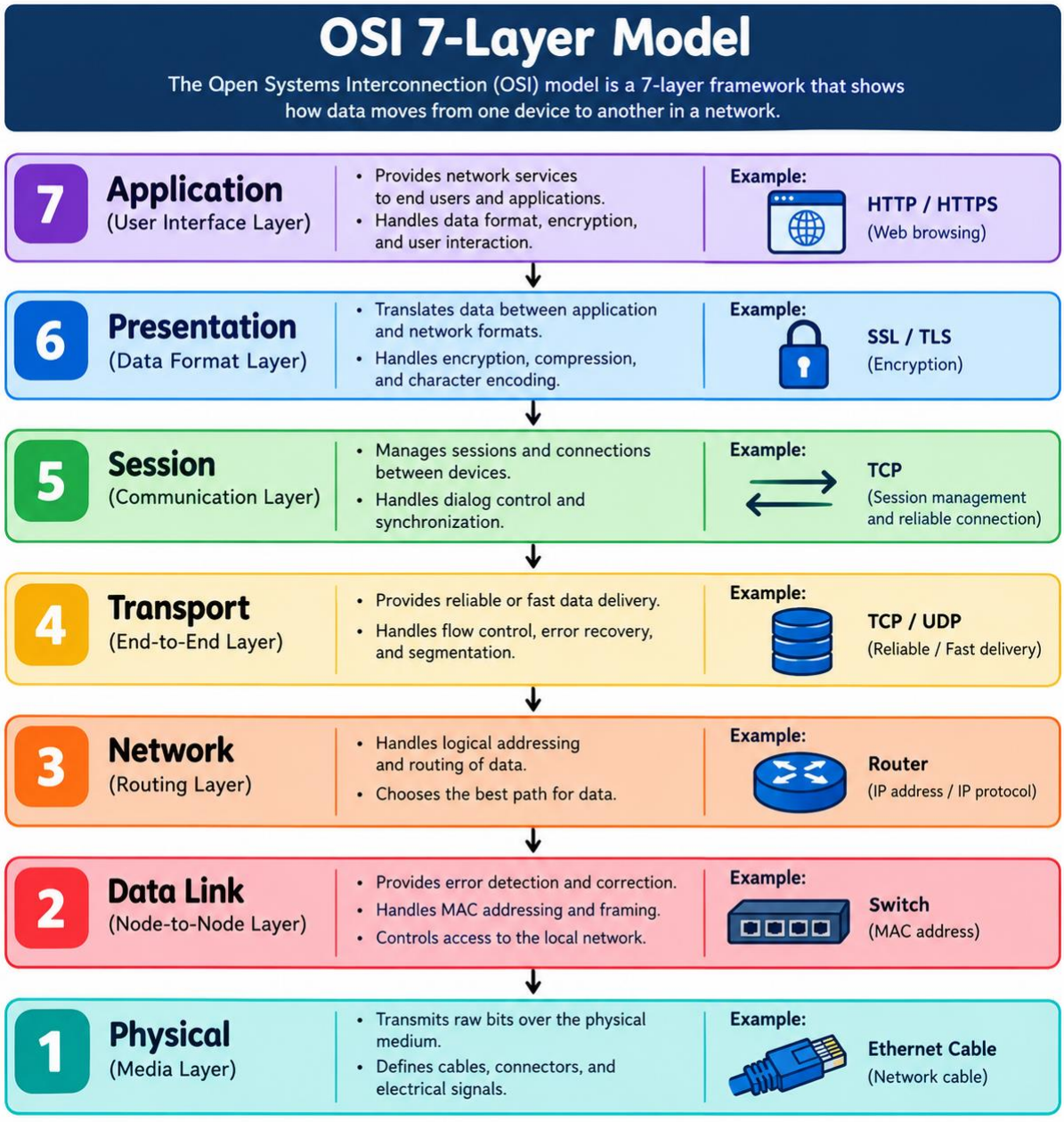
OSI Layer

Task 1:- Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



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Task 2:- Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames. Write your answers as 'Scenario: Layer Name (Layer Number)'.

a) Encrypting data before sending: Presentation (Layer 6)

b) Routing packets across networks: Network (Layer 3)

c) Converting data to electrical signals: Physical (Layer 1)

d) Opening a web page in your browser: Application (Layer 7)

e) Detecting and correcting errors in frames: Data Link (Layer 2)

Task 3:- Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

1. Application Layer (Layer 7): WhatsApp allows me to type and send messages, photos, and videos.

2. Transport Layer (Layer 4): TCP or other transport mechanisms help transfer data between devices reliably, depending on the connection.

3. Network Layer (Layer 3): IP addresses help route data from my phone across networks to the destination.

4. Data Link Layer (Layer 2): Wi-Fi or mobile network technology transfers data across the local network link.

5. Physical Layer (Layer 1): Radio signals carry data between my phone and the Wi-Fi router or mobile tower.

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Task 4:- Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

1. Application Layer (Layer 7): The website's HTTP/HTTPS service may be down, or there may be a browser or website configuration issue.

2. Transport Layer (Layer 4): The TCP connection may be blocked or failing because ports 80 (HTTP) or 443 (HTTPS) are unavailable or blocked by a firewall.

Reasoning: Since you can successfully ping the website's IP address, basic IP connectivity is working. However, ping uses ICMP, not HTTP or HTTPS, so a successful ping does not guarantee that the website itself is accessible.